ORACLE DBA

Interview Questions & Answers

Covers: Patching, Backup & Recovery, RAC & Cluster, and Performance Tuning. Questions that ask about your own experience (for example "Have you worked on...") have model answers; adapt them to what you have actually done. Commands are examples; adjust paths, names and versions for your environment.

- 1. Patching (10 questions)
- 2. Backup & Recovery (7 questions)
- 3. RAC & Cluster (6 questions)
- 4. Performance Tuning (4 questions)

1. Patching

Q1.1 What is database patching?

Database patching is the process of applying fixes and updates supplied by Oracle to the Oracle software (the binaries in ORACLE_HOME) and, where required, to the database itself (data dictionary / SQL changes).

Its goals are to fix bugs, close security vulnerabilities and improve stability. Patching is done with the OPatch utility (or opatchauto for Grid Infrastructure/RAC), followed by running datapatch on the database for SQL-level changes (12c and later).

Q1.2 What is an Oracle patch?

An Oracle patch is a collection of files and scripts released by Oracle to fix one or more specific bugs, deliver security fixes or add small enhancements to an existing installation.

Each patch has a unique patch number on My Oracle Support (MOS) and a README that explains the prerequisites, the apply steps and the post-install steps.

Q1.3 Why is patching important?

- **Security:** Critical Patch Updates (CPU) close known vulnerabilities.
- **Stability:** fixes known bugs that can cause crashes, hangs or wrong results.
- **Performance:** many patches include optimizer and performance fixes.
- **Support and compliance:** Oracle Support expects a supported patch level, and audits often require current CPUs.
- **Compatibility:** keeps the database aligned with OS, Grid and application requirements.

Q1.4 What are the different types of Oracle patches?

- **Interim (one-off) patch:** fixes a single bug, often provided for a specific customer issue.
- **Merge patch (MLR):** combines several one-off patches into one to avoid conflicts.
- **Bundle patch (BP):** a cumulative set of fixes for a product or platform (for example Windows or Exadata bundle patches).
- **CPU (Critical Patch Update):** quarterly security-only fixes.
- **PSU (Patch Set Update):** quarterly cumulative patch containing the CPU plus important bug fixes (11g and 12.1).
- **Release Update (RU) / Release Update Revision (RUR):** the quarterly replacement for PSUs from 12.2 and 18c onwards.
- **Patch Set:** a full release such as 11.2.0.3 to 11.2.0.4, installed as an upgrade.
- **Diagnostic patch:** collects extra trace/diagnostic information for Oracle Support.

Q1.5 What is PSU patching in Oracle?

A Patch Set Update (PSU) is a quarterly, cumulative, well-tested patch bundle. It contains the security fixes of the Critical Patch Update plus a set of high-impact, low-risk bug fixes. Because it is cumulative, you only need the latest PSU. It increments the 5th digit of the version, for example 11.2.0.4.200114.

Typical steps:

- Download the PSU and the latest OPatch; back up ORACLE_HOME and the inventory.
- Shut down the database and listener, run opatch apply, then restart.
- Run the post-install SQL: catbundle.sql psu apply (11g) or datapatch -verbose (12c and later).
- Verify the result with opatch lsinventory and the dictionary views.

Q1.6 How do you know if a PSU patch is installed in a database?

Check both the software inventory and the database registry:

```
$ORACLE_HOME/OPatch/opatch lspatches
$ORACLE_HOME/OPatch/opatch lsinventory -detail
SQL> select patch_id, action, status, description from dba_registry_sqlpatch; -- 12c and
later
SQL> select action_time, action, version, comments from dba_registry_history; -- 11g
```

opatch shows what is applied to the binaries; dba_registry_sqlpatch / dba_registry_history show whether the SQL part has been applied inside the database.

Q1.7 How do you know if you have an OPatch conflict?

Run the conflict pre-check before applying the patch. It compares the new patch against everything already in the home:

```
cd <patch_directory>
$ORACLE_HOME/OPatch/opatch prereq CheckConflictAgainstOHWithDetail -ph ./
```

opatch apply also runs this check automatically and stops if a conflict exists. To resolve a conflict, roll back the conflicting one-off patch if it is no longer needed, or request a merge patch from Oracle Support (MOS) that covers both fixes.

Q1.8 Have you applied OCM patching?

OCM is Oracle Configuration Manager. In older OPatch versions (11.2 and 12.1), OPatch asked for an OCM response file when applying a patch. You generate it once and pass it on the command line:

```
$ORACLE_HOME/OPatch/ocm/bin/emocmrsp
$ORACLE_HOME/OPatch/opatch apply -ocmrf /path/ocm.rsp
```

You can create the response file without registering an email address. Later OPatch releases dropped the OCM requirement, so a recent OPatch does not ask for it.

(This is a model answer; describe your own hands-on experience in the interview.)

Q1.9 How do you patch if OUI / inventory is not available?

OPatch needs a valid central inventory. If the inventory is missing or corrupt:

- Find the inventory location in /etc/oraInst.loc (or oraInst.loc in the inventory pointer file).
- Restore the inventory from backup if you have one, or re-attach the Oracle home:

```
cd $ORACLE_HOME/oui/bin
./runInstaller -silent -attachHome ORACLE_HOME="/u01/app/oracle/product/19c/dbhome_1"
ORACLE_HOME_NAME="OraDb19c_home1"
```

- Verify with opatch lsinventory, then apply the patch.
- If a different inventory must be used, point to it with the -invPtrLoc option of opatch.

Q1.10 If you have successfully patched a 3 node RAC environment, the cluster status is showing as 'rolling patch'. Why is it showing as rolling patch?

During rolling patching, the nodes run at different patch levels for a while. Clusterware/ASM therefore marks the cluster state as ROLLING PATCH to allow the mixed-version situation. It returns to NORMAL when every node has the same patch level.

If it still shows rolling patch after all three nodes are done, then:

- one node was missed, or has a different patch level; or
- the post-patch step was not completed on one node (rootcrs.sh -postpatch), or the stack there was not restarted.

Check and fix:

```
crsctl query crs activeversion -f
crsctl query crs releasepatch
crsctl query crs softwarepatch
asmcmd showclusterstate
SQL> select sys_context('sys_cluster_properties','cluster_state') from dual;
```

Compare the patch level on every node (kfod op=patches / opatch lspatches), patch the node that differs, run rootcrs.sh -postpatch where it was not run, and re-check the state.

2. Backup & Recovery

Q2.1 What is a Recovery Catalog?

A recovery catalog is a schema, normally in a separate database, that stores RMAN repository metadata: backup sets and pieces, datafile copies, archived logs, the database structure and RMAN stored scripts.

Without a catalog, RMAN keeps this information only in the target database control file (limited by CONTROL_FILE_RECORD_KEEP_TIME). A catalog gives:

- longer retention of backup history than the control file
- stored scripts and centralized reporting for many databases
- easier restore when the control file is lost

Setup:

```
SQL> create user rcat identified by ***** default tablespace rcat_ts quota unlimited on
rcat_ts;
SQL> grant recovery_catalog_owner to rcat;
RMAN> connect catalog rcat/*****@catdb
RMAN> create catalog;
RMAN> connect target /
RMAN> register database;
```

Q2.2 What is the difference between Media Recovery and Crash Recovery?

Crash recovery is automatic. It happens at startup after an instance failure (power loss, shutdown abort, server crash). SMON uses the online redo logs to roll forward committed and uncommitted changes, then rolls back uncommitted transactions using undo. It needs no backup and no DBA action.

Media recovery is needed when files are lost or damaged (datafile, control file, disk failure). The DBA must restore the files from backup and apply archived and online redo logs. It can be complete or incomplete (point-in-time), and it requires the database to be in ARCHIVELOG mode.

Q2.3 What is the difference between Restoring a database and Recovering a database?

Restore means copying the backup files (datafiles, control file, spfile) from the backup location back to disk. The files are then as old as the backup.

Recover means applying redo (archived logs, online redo logs) and incremental backups to the restored files to bring them forward to the current time or to a chosen SCN/time.

In RMAN you run RESTORE DATABASE first and then RECOVER DATABASE.

Q2.4 How do you resolve a thousand archive log gap in a standby database?

First confirm the gap on the standby:

```
SQL> select * from v$archive_gap;
SQL> select thread#, max(sequence#) from v$archived_log where applied='YES' group by
thread#;
```

Then choose the approach:

- **Archive logs still exist on the primary (or in backup):** copy or restore them to the standby and register them (ALTER DATABASE REGISTER LOGFILE '<file>'), or let FAL_SERVER/FAL_CLIENT fetch them. Then resume managed recovery.
- **Logs were deleted or the gap is too large:** roll the standby forward with an RMAN incremental backup from the standby's current SCN.

12c and later (one command from the standby):

```
RMAN> recover database from service <primary_service> noredo using compressed backupset;
```

11g (manual):

```
SQL> select current_scn from v$database; -- on standby, with MRP stopped
RMAN> backup incremental from scn <scn> database format '/tmp/inc_%U'; -- on primary
RMAN> catalog start with '/tmp/'; -- on standby
RMAN> recover database noredo;
```

Afterwards refresh the standby control file if datafiles were added, restart managed recovery and check that no new gap appears. Also find out why the gap happened (network, space, defer state, listener).

Q2.5 How to avoid block corruption in standby database?

- Enable FORCE LOGGING on the primary, because NOLOGGING operations leave unlogged blocks that become logically corrupt on the standby.
- Set DB_BLOCK_CHECKSUM=TYPICAL (or FULL) and DB_BLOCK_CHECKING=MEDIUM/FULL on primary and standby (DB_ULTRA_SAFE sets these).
- Set DB_LOST_WRITE_PROTECT=TYPICAL on both so lost writes are detected.
- Use Active Data Guard, which repairs corrupt blocks automatically from the other side.
- Use ASM redundancy and reliable storage.
- Run RMAN backup validate check logical database regularly and monitor v$database_block_corruption and the alert log.

Q2.6 What will you check if archive logs are not arriving at standby database?

- **Primary:** v$archive_dest and v$archive_dest_status for errors; LOG_ARCHIVE_DEST_n is valid and its state is ENABLE (not DEFER); LOG_ARCHIVE_CONFIG has the right DG_CONFIG.
- **Network:** tnsping to the standby, listener running on the standby, correct tnsnames/service name, firewall open.
- **Authentication:** the password file is the same on both sides and REMOTE_LOGIN_PASSWORDFILE=EXCLUSIVE.
- **Standby state:** database is mounted, RFS and MRP processes are running (v$managed_standby), standby redo logs exist for real-time apply.
- **Space:** enough free space in the standby archive destination / FRA.
- **Settings:** FAL_SERVER and FAL_CLIENT are set; LOG_ARCHIVE_MAX_PROCESSES is adequate.
- **Logs:** alert logs on both servers and v$dataguard_status.

Q2.7 What will you do if the archive log destination runs out of storage during a peak transaction period, causing the database to halt?

Archiving stops (ORA-00257 / archiver error) and the instance hangs until space is available. Fix it in this order:

- Check usage: v$recovery_file_dest and v$flash_recovery_area_usage.
- Free space safely with RMAN, not with OS rm:

```
RMAN> backup archivelog all delete input;
RMAN> delete archivelog all completed before 'sysdate-1';
```

- If an OS delete happened, run crosscheck archivelog all and delete expired archivelog all.
- Or add space: increase db_recovery_file_dest_size, add disk, or point log_archive_dest_1 to a location with free space.

```
SQL> alter system set db_recovery_file_dest_size=200G scope=both;
```

The database resumes automatically once space is available.

Make sure archive logs needed by the standby are not deleted (use CONFIGURE ARCHIVELOG DELETION POLICY TO APPLIED ON STANDBY).

Prevention: alert at 70 to 80 percent usage, back up archive logs every 15 to 30 minutes, set a deletion policy, and size the FRA for peak load.

3. RAC & Cluster

Q3.1 What is the Cache Fusion mechanism in RAC?

Cache Fusion lets RAC instances share data blocks directly over the private interconnect from one instance's buffer cache to another's, instead of writing the block to disk and having the other instance read it back (the old "block pinging").

It is implemented by the Global Cache Service (GCS) and Global Enqueue Service (GES), using background processes such as LMS, LMD, LMON and LCK. The Global Resource Directory (GRD) tracks which instance holds each block and in which mode (shared, exclusive, consistent read). The result is that all instances behave as one shared cache.

Q3.2 What is the SCAN Listener and what is the purpose of SCAN Listener?

SCAN (Single Client Access Name), introduced in 11gR2, is one hostname for the whole cluster. It resolves, via DNS (round robin) or GNS, to up to three SCAN VIPs. A SCAN listener runs on each SCAN VIP.

Purpose:

- Clients connect to the SCAN name only, so connection strings do not change when nodes are added or removed.
- The SCAN listener load-balances and redirects the client to the local listener of the least-loaded instance offering the service.
- It gives simpler failover and administration.

Q3.3 How do you locate SCAN Listener across the nodes?

Run these as the Grid Infrastructure owner:

```
srvctl config scan
srvctl status scan
srvctl config scan_listener
srvctl status scan_listener
crsctl stat res -t | grep -i scan
ps -ef | grep tnslsnr | grep -i scan
lsnrctl status LISTENER_SCAN1
```

srvctl status scan_listener shows which node each SCAN listener is running on. lsnrctl must use the Grid home environment.

Q3.4 What is ASM Storage Architecture?

Automatic Storage Management (ASM) is Oracle's volume manager and file system for database files.

- **ASM instance (+ASM):** a small instance on each node that manages disk groups and metadata (processes include RBAL, ARBn, GMON). Database instances talk to it through ASMB.
- **Disk group:** a collection of ASM disks (LUNs or partitions) managed as one unit.
- **Allocation unit and extents:** files are striped in allocation units (1 MB by default) across all disks in the group.
- **Redundancy:** external, normal (two-way mirror) or high (three-way mirror), using failure groups.
- **Rebalance:** adding or dropping disks automatically redistributes data.
- **Extras:** ACFS (ASM Cluster File System), ADVM volumes, and Flex ASM (12c), where ASM instances need not run on every node.
- Typical use: datafiles, redo, control files, spfile, OCR and voting files.

Tools: asmcmd, v$asm_diskgroup, v$asm_disk, v$asm_operation.

Q3.5 Have you worked on ASM Storage Architecture?

Model answer (adapt it to your own experience): Yes. I have created and managed disk groups, added and dropped disks and monitored the rebalance (v$asm_operation), checked free space with v$asm_diskgroup and asmcmd lsdg, handled disks with asmcmd/ASMLib/udev, moved databases onto ASM with RMAN and worked with the OCR and voting file disk groups in RAC.

Q3.6 What is Oracle Clusterware status 'rolling patch' after patching a 3 node RAC?

The status is "In Rolling Patch" (cluster upgrade state ROLLING PATCH). It is a normal, temporary state while nodes are patched one at a time and hold different patch levels. Once every node is patched, it should return to NORMAL.

If it stays in rolling patch, a node is still at a different patch level or its post-patch step did not complete. Check:

```
crsctl query crs activeversion -f
asmcmd showclusterstate
```

Patch the missing node, run rootcrs.sh -postpatch if needed and re-check (see the same question in the Patching section).

4. Performance Tuning

Q4.1 In an AWR report, what are the things we need to check?

- **Header:** snapshot interval, elapsed time and DB time (DB time divided by elapsed time gives average active sessions), CPU count, and RAC/instance details.
- **Load Profile:** DB time, DB CPU, redo size, logical and physical reads, parses and hard parses, executes, transactions per second.
- **Instance Efficiency:** buffer hit, library cache hit, soft parse %, parse CPU to parse elapsed.
- **Top Timed Foreground Events:** what the database waits on (DB CPU, db file sequential/scattered read, log file sync, enq contention, etc.).
- **SQL Statistics:** SQL ordered by elapsed time, CPU, gets, reads, executions and parse calls.
- **Time Model, Wait Event classes, Segments by reads/waits, Latch and Enqueue activity.**
- **Memory advisories:** SGA, PGA, buffer cache and shared pool; plus OS/host CPU and I/O statistics.
- **RAC:** interconnect and global cache statistics.

Compare against a baseline (AWR compare periods report) from a good day.

Q4.2 What are the different tools provided by Oracle to assist the performance tuning?

- **AWR** (Automatic Workload Repository) and **Statspack** for historical statistics.
- **ADDM** (Automatic Database Diagnostic Monitor) for automatic analysis and recommendations.
- **ASH** (Active Session History) for session-level history.
- **SQL Trace / TKPROF** and **EXPLAIN PLAN / DBMS_XPLAN** for individual statements.
- **SQL Tuning Advisor**, **SQL Access Advisor** and **SQL Plan Management** (baselines, profiles).
- **Real-Time SQL Monitoring** for long-running statements.
- **Segment, SGA, PGA and buffer cache advisors.**
- **Enterprise Manager / Cloud Control** and dynamic performance views (v$session, v$sql, v$system_event).
- **OSWatcher** for OS-level data and DBMS_STATS for optimizer statistics.

Q4.3 Your production database is suddenly slow during peak hours. Users are complaining about slow response time and got the error 'Application timed out'. What are the steps for resolving this issue?

1. Scope the problem. Ask when it started, which module or users, and what changed (deployment, data load, patch, stats job).

2. Check the server. CPU, memory/swap, I/O and load with top, vmstat and iostat.

3. Check what sessions are doing now. Use v$session (event, sql_id, blocking_session), v$active_session_history, blocking locks and long transactions:

```
SQL> select sid, serial#, event, sql_id, blocking_session, seconds_in_wait from v$session
where status='ACTIVE' and type='USER';
```

4. Review the alert log for errors such as ORA-00020 (max processes), ORA-00257, ORA-04031 or deadlocks, and check the listener log for a connection storm.

5. Generate AWR / ASH / ADDM reports for the peak window and compare with a normal period.

6. Find the top SQL and waits. Look for plan changes, stale or missing statistics, full scans, high parse counts, log file sync, buffer busy waits or latch/library cache contention. Run the SQL Tuning Advisor.

7. Mitigate immediately. With approval, kill the blocking or runaway sessions, fix the plan (SQL profile/baseline, gather stats), add an index, or add resources (SGA, connections, temp/undo).

8. Check the application tier. Connection pool sizing and application timeout settings.

9. Prevent recurrence. Baselines, proactive monitoring and alerts, capacity planning and regular statistics maintenance.

Q4.4 Why do we need to rebuild an index?

Common reasons:

- **Unusable index:** after ALTER TABLE ... MOVE, partition maintenance or a failed load, the index becomes UNUSABLE and must be rebuilt.
- **Heavy fragmentation:** many deletes and updates leave sparse blocks and deleted leaf entries, which wastes space and increases I/O.
- **Index height or space problems:** a very high index height, or an index much larger than the data justifies.
- **Move or change:** to move an index to another tablespace or change its storage attributes.
- **Corruption:** to recreate a damaged index.

Check before rebuilding:

```
SQL> analyze index <idx> validate structure;
SQL> select height, lf_rows, del_lf_rows from index_stats;
```

Rebuild only when del_lf_rows is a large share of lf_rows (for example over 20 percent) or the height is excessive. Oracle does not recommend routine rebuilds of healthy B-tree indexes.

```
SQL> alter index <idx> rebuild online;
SQL> alter index <idx> coalesce; -- lighter alternative
```